UBC Math 101 Final Exam 150 minutes

24 April, 2025

- The test consists of 28 pages and 10 questions, worth a total of 100 marks.
- This is a closed-book examination. Resources that you are not allowed to use include, but are not limited to: notes, cell phones, calculators, collaboration, looking at another student's exam, smart watches, and internet access. Students found engaging in misconduct will be reported.
- You may not share the content of this exam with anyone else, in person or online, now or in the future. You may not discuss the content of this exam with anyone before they take it.
- Fill in the information below before turning to the questions. No work on this page will be marked. You must stop writing promptly when the end of the test is announced. Failure to do so may result in a grade penalty and a report of academic misconduct.
- Show your work unless otherwise instructed. A correct answer with no supporting work may receive no credit.

Student number								
Section								
Name	.....							
Signature								

Additional instructions

- Please use the spaces indicated. You may write in pen or pencil, but write clearly, darkly, and in the appropriate spaces. Clearly $\boxed{\text{box}}$ your final answers.
- Keep away from the corners and extreme margins of the page, because your work will be cut and scanned and the edges may not be recorded. Do not dismember your exam. Do not mark over any QR codes.
- If you require extra paper, raise your hand.
  - You must put your name and student number at the centre top of any extra pages. Don't put information in corners, as these may be cut off.
  - You must indicate the test-number and question-number.
  - Please do this **on both sides** of any extra pages.
- The purpose of this exam is for you to demonstrate your understanding of course concepts. Keep this in mind when deciding exactly what you want to write down.
- On simplification: pursuant to the previous bullet point, aggressively refusing to simplify things that should be obvious (for example, keeping a term as '$\sin \pi$' without changing it to '0') may lead graders to flag a lack of understanding. If you are not asked to simplify, however, there is no need to carry out more time-intensive calculations. For example, you might leave $\frac{100}{55^2}$ as it is.

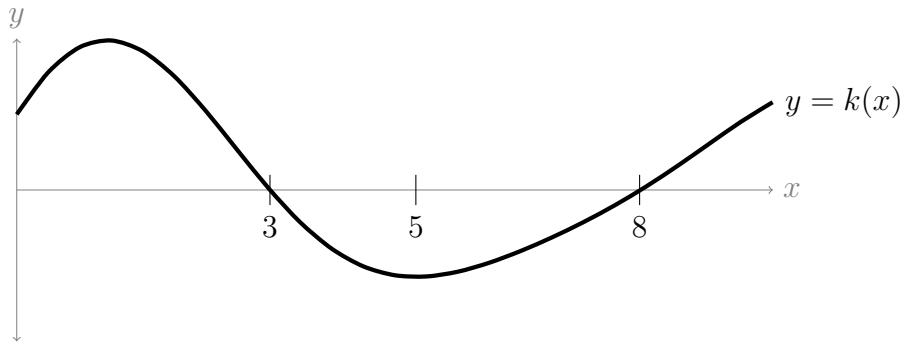
1. (a) [4 marks] Let $f(x)$ and $g(x)$ be integrable functions. If

$$\int_3^5 f(x)\,\mathrm{d}x = 2, \qquad \int_0^5 f(x)\,\mathrm{d}x = 3, \qquad \text{and} \qquad \int_0^3 g(x)\,\mathrm{d}x = 7$$

then find and simplify $\displaystyle\int_0^3 (f(x) + g(x))\,\mathrm{d}x$.

(b) [3 marks] For the function $k(x)$ whose graph is given below, list the following quantities in increasing order, from smallest to largest.

- $A = \displaystyle\int_0^3 k(x)\,\mathrm{d}x$
- $B = \displaystyle\int_0^5 k(x)\,\mathrm{d}x$
- $C = \displaystyle\int_0^8 k(x)\,\mathrm{d}x$



Answer: ____ < ____ < ____

(c) [2 marks] A certain population of salmon swims upstream, past a research station. A biologist keeps track of how many individual salmon pass the station over time. Suppose $\int_a^b q(t)\,\mathrm{d}t$ gives the number of salmon that passed the station from time $t = a$ to time $t = b$, where t is measured in hours.

Select the best interpretation of $q(t)$ from the list below.

A. $q(t)$ gives the total distance (measured in kilometres) the salmon have travelled from time $t = a$ to time $t = b$.

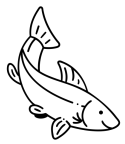
B. $q(t)$ gives the acceleration of the salmon, from time $t = a$ to time $t = b$.

C. $q(t)$ gives the number of salmon that passed the station from time $t = a$ to time $t = b$, where t is measured in hours.

D. $q(t)$ gives the rate (in number of salmon per hour) at which salmon pass the research station at time t.

E. $q(t)$ gives the rate (in kilometres per hour) at which salmon pass the research station at time t.

F. $q(t)$ gives the rate of change (in salmon per hour per hour, $\frac{\text{salmon}}{\text{hour}^2}$) of the rate at which salmon are passing the research station.



2. Compute each integral. Simplify your final answers.

(a) [4 marks] $\displaystyle\int \cos x\sqrt{\sin x}\,\mathrm{d}x$

Question continues on other side

(b) 6 marks $\displaystyle\int \frac{5x+8}{x^2+3x+2}\,\mathrm{d}x$

(c) 6 marks $\displaystyle\int \frac{1}{x^2\sqrt{1+x^2}}\,\mathrm{d}x$

 *Extra space to continue your solution to* ***Question 2(c)****.*

3. A continuous random variable X has probability density function (PDF)

$$f(x) = \begin{cases} 2x & \text{if } 0 \leq x \leq 1 \ , \\ 0 & \text{otherwise} \ . \end{cases}$$

(a) [3 marks] Imagine the trial being repeated many times. In what percentage of trials do you expect to find $X \leq \frac{1}{2}$?
(Make sure to give your answer as a percentage, i.e. a number between 0 and 100.)

(b) [4 marks] You may assume, without proof, that $\mathbb{E}(X) = \frac{2}{3}$. Compute $\sigma(X)$.

 *Extra space to continue your solution to* ***Question 3****.*

4. [6 marks] Find the radius of convergence of the series

$$\sum_{n=1}^{\infty} \frac{4^n}{\sqrt{n}}(x-6)^n.$$

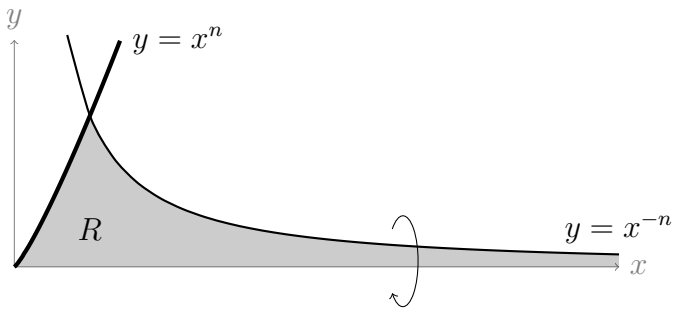
 *Extra space to continue your solution to* ***Question 4****.*

5. [10 marks] Use Taylor series to find the following limit:

$$\lim_{x \to 0} \frac{e^{x^4} - x^2 \sin(x^2) - 1}{x^8}.$$

 *Extra space to continue your solution to* ***Question 5****.*

6. [8 marks] Consider the region R in the first quadrant bounded by the x-axis and the curves $y = x^{-n}$ and $y = x^n$, where $n > 1$ is a real number.



Find the volume $V(n)$ of the solid generated by revolving R about the x-axis. Express your answer in terms of n.

 *Extra space to continue your solution to* ***Question 6****.*

7. A small rocket propels itself straight upwards. From $t = 0$ to $t = 4$ minutes, its velocity is given by:

$$v(t) = \frac{100}{1 + t^3} \quad \text{meters/min.}$$

(a) [5 marks] Estimate the height the rocket gains from $t = 0$ to $t = 4$ using the trapezoid rule with 4 trapezoids.

(You may leave your sum in calculator-ready form.)

(b) [5 marks] When we were calculating the error of our approximation, we calculated that

$$|v''(t)| = \left| 600t \frac{1 - 2t^3}{(1 + t^3)^3} \right| \leq 307200 \ (= 64 \cdot 4800),$$

for all $0 \leq t \leq 4$, which is too much. But we are dedicated and want to make our estimation work by using more trapezoids. To make our error at most 64 meters, using the bound given above, how many trapezoids would we need? Simplify your answer.

 *Extra space to continue your solution to* ***Question 7****.*

8. In this question, a is a real number.

(a) [4 marks] Find, with justification, all values of a for which the series

$$\sum_{n=1}^{\infty} e^{(n^2)} e^{-an}$$

converges.

Find, with justification, all values of a for which the series

$$\sum_{n=1}^{\infty} n e^{-an}$$

converges.



Extra space to continue your solution to ***Question 8(a),(b)****.*

(b) [6 marks] Find, with justification, all values of a for which the series

$$\sum_{n=1}^{\infty} e^n e^{-an}$$

converges.

Evaluate the series (in terms of a) for all values of a for which it converges.

 *Extra space to continue your solution to* ***Question 8(c)****.*

9. [10 marks] Let a and b be real numbers and let

$$f(x) = \begin{cases} 0 & \text{if} \quad x < a \ , \\ e^{bx} & \text{if} \quad x \geq a \ . \end{cases}$$

Find (with justification) the values for a and b such that $f(x)$ is a probability density function with expected value equal to 0.

 *Extra space to continue your solution to* ***Question 9****.*

10. [8 marks] Let

$$A(x) = \int_x^{x^2} e^{-x} e^{3t} \, \mathrm{d}t.$$

Find $A'(x)$.

Note: the presence of both x and t in the integrand is not a typo.

 *Extra space to continue your solution to* ***Question 10****.*

Formula Sheet

You may detach this page, but it must be turned in with your exam.

Error in Numerical Approximations

Suppose you are approximating the integral $\int_a^b f(x)\mathrm{d}x$.

- Assume that $|f''(x)| \leq M$ for all $a \leq x \leq b$. Then the total error introduced by the trapezoidal rule is bounded by $\frac{M}{12}\frac{(b-a)^3}{n^2}$.
- Assume that $|f^{(4)}(x)| \leq L$ for all $a \leq x \leq b$. Then the total error introduced by Simpson's rule is bounded by $\frac{L}{180}\frac{(b-a)^5}{n^4}$.

Common Taylor Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots \quad \text{for all } -\infty < x < \infty$$

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)!}x^{2n+1} = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \cdots \quad \text{for all } -\infty < x < \infty$$

$$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n)!}x^{2n} = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \cdots \quad \text{for all } -\infty < x < \infty$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots \quad \text{for all } -1 < x < 1$$

$$\log(1+x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \quad \text{for all } -1 < x \leq 1$$

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots \quad \text{for all } -1 \leq x \leq 1$$

Taylor Remainder

The Lagrange Remainder Formula guarantees that for any evaluation point x, if M_{n+1} is a constant such that $|f^{(n+1)}(c)| \leq M_{n+1}$ for all c between x and a, then

$$|E_n| = |f(x) - T_n(x)| \leq \frac{M_{n+1}}{(n+1)!}|x-a|^{n+1}.$$

*scrap paper – **nothing** on this page will be graded*
You may detach this sheet, but make sure you turn it in with your exam.

THIS PAGE IS NOT VISIBLE TO GRADERS
NOTHING HERE WILL BE MARKED